RR-F-183
AUGUST 13, 1946

FEDERAL SPECIFICATION

FENCE-POSTS, GATES, AND ACCESSORIES

This specification was approved on the above date by the Director of Procurement, for the use of all departments and establishments of the Government, and shall become effective not later than November 1, 1946, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A–1. The following Federal Specifications, of the issue in effect on date of invitation for bids, shall form a part of this specification:

QQ–M–151—Metals: General Specification for Inspection of.
QQ–Z–351—Zinc; Slab (Spelter).
RR–F–191—Fencing; Chain-Link or Welded.

B. GRADE

B–1. Fence accessories in accordance with this specification shall be of one grade and shall consist of the following, as specified in the invitation for bids:

a.—Gates (single and double).
b.—Gate posts.
c.—Corner, end, and pull posts.
d.—Line posts.
e.—Post braces.
f.—Post tops.
g.—Barbed-wire supporting arms.
h.—Top rails.
i.—Stretcher bars and clips.
j.—Top and bottom reinforcing wires.

C. MATERIALS AND WORKMANSHIP

C–1. *Materials.*—Posts, gate frames, braces, rails, stretcher bars, and truss rods shall be of steel having a carbon content of not less than 0.10 percent; reinforcing wires shall be of high carbon steel; and gate hinges, post caps, barbed-wire supporting arms, stretcher bar bands, and other parts shall be of steel, wrought iron, or malleable iron, except that fabric bands, clips, bolts, and nuts may be of aluminum.

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C–2. *Workmanship* shall be first class. Parts shall be formed accurately to dimension. Zinc-coated surfaces shall be free from imperfectly coated spots, bruised or scaled coating, drops of zinc, and sal ammoniac spots.

D. GENERAL REQUIREMENTS

D–1. Posts, gate frames, rails, and braces shall conform to the dimensions and weights shown in table I.

TABLE I.—*Dimensions and weights*

Use and section	Outside diameter or dimensions, minimum	Weight per foot, minimum
	Inches	*Pounds*
End, corner, and pull posts (tubular) for fabric heights:		
6 feet and less: Round	2.375	3.65
Square	2.0	3.85
Over 6 feet: Round	2.875	5.79
Square	2.5	5.70
Gate posts for nominal width of gate, single, or one leaf of double, for fabric heights:		
6 feet and less: Round	2.875	5.79
Square	2.5	5.70
Over 6 feet to 13 feet, incl.		
Round	4.0	9.10
Square	3.0	9.10
Over 13 feet to 18 feet, incl.		
Round	6.625	18.97
Over 18 feet: Round	8.625	24.70
Gates; exterior frames for fabric heights:		
6 feet and less: Round	1.625	2.27
Square	1.50	1.90
Over 6 feet: Round	1.875	2.71
Square	2.00	2.60
Interior bracing (tubular):		
Round	1.625	2.27
Square	1.50	1.90
Rails and post braces (tubular):		
Round	1.625	2.27
Square	1.50	1.90
Intermediate posts for fabric heights:		
6 feet and less:		
Tubular (round)	1.875	2.71
U-section	2.125x1.875x$\frac{1}{4}$	2.65
H-section	1.875x1.625x$\frac{1}{8}$	2.60
Over 6 feet:		
Tubular (round)	2.375	3.65
U-section	2.5x1.875x$\frac{1}{4}$	4.00
U-section (modified with flanges)	.....	3.65
H-section	2.25x1.95x9/64	4.10

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D–2. *Zinc coating.*—All steel and iron parts shall be zinc-coated after fabrication, using zinc grade E in accordance with Federal Specification QQ–Z–351. The zinc coating shall withstand six 1-minute dips of the Preece test. (See par. F–3.)

D–2a. *Special finishes.*—When so specified, all zinc-coated material, after completion of the coating process, shall be treated by a dip process which shall prevent the formation of white corrosion products when test specimens are subjected to a 20 percent salt spray solution for 100 hours at 95° F. in a standard salt spray cabinet. The process shall provide a finish suitable for the application of paint. When so specified, the first dip process shall be followed by a second dip process which shall provide a permanent color as specified.

E. DETAIL REQUIREMENTS

E–1. *Gates* shall be swing or sliding as specified, complete with latches, stops, keepers, hinges, or rollers and roller tracks, and fabric, and, when so specified, with locks and provision for three strands of barbed wire above the fabric.

E–1a. *Gate frames* shall be constructed of tubular members and, for gate leaves more than 8 feet wide, shall have intermediate members and/or diagonal truss rods as necessary to provide rigid construction of ample strength and free from sag and twist. Where a barbed wire top is specified, the end members of gate frames shall be extended approximately 1 foot above the top member and arranged for attaching three uniformly spaced strands of barbed wire and furnished with clips or other suitable method for securely attaching the wire. All joints between tubular members shall be made by welding or by means of heavy fittings providing rigid and watertight connections. Truss rods shall be not less than ⅜ inch in nominal diameter.

E–1b. *Fabric* shall conform to Federal Specification RR–F–191 and, unless specified otherwise, shall be type A, zinc-coated, class 1, chain-link, 2-inch mesh of 0.148-inch diameter wire. The fabric shall be attached securely to the gate frame at intervals not exceeding 15 inches.

E–1c. *Hinges* shall be of heavy pattern, of adequate strength for the gate, and with large bearing surfaces for clamping in position. The hinges shall not twist or turn under the action of the gate. The gates shall be capable of being opened and closed easily by one person and shall be arranged so they may be swung back parallel with the fence.

E–1d. *Latches, stops, and keepers* shall be provided for all gates. Latches shall be of the plunger-bar type and shall be of the full gate height and arranged to engage the gate stop, except that for single gates for openings less than 10 feet wide a forked latch may be provided. Latches shall be arranged for locking. Stops shall consist of a flush plate with anchor arranged to be set in concrete and to engage the plunger of the bar latch, except that for single gates for openings less than 10 feet wide other approved types of stops may be provided. Keepers shall consist of a substantial mechanical device for securing and supporting the free end of the gate when in the full open position.

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E–1e. *Lock*, when specified to be furnished, shall be of substantial construction and shall be accessible from both sides of the gate. Three keys shall be furnished with each lock.

E–2. *Posts* shall be of the lengths specified and shall be tubular, except that line posts, in lieu of tubular, may be H–beam or U sections. Line posts shall be used only in straight runs between gate, corner, pull, and end posts. (See par. I–5.)

E–3. *Post braces* shall be provided for each gate, corner, pull, and end post for use with fabric 5 feet or more in height, and shall consist of a round tubular brace extending to each adjacent line post at midheight of the fabric, and a truss consisting of a double 0.192-inch-nominal-diameter wire, a rod not less than ⅜ inch in nominal diameter, or equivalent, from the line post back to the gate, corner, pull, or end post, with a turnbuckle or other equivalent provision for adjustment.

E–4. *Post tops* shall consist of ornamental caps or combination caps and barbed-wire supporting arms, as specified. When so specified or when a top rail is to be provided, the post shall be provided with a hole suitable for the through passage of the top rail. The post tops shall fit snugly on the posts, shall have suitable means for being attached securely, and shall exclude moisture from tubular posts.

E–4a. *Barbed-wire supporting arms*, when specified to be furnished, shall be at an angle of approximately 45°, and shall be fitted with clips or other suitable means for attaching three lines of barbed wire, with the top outside wire approximately 12 inches horizontally from the fence line and the other wires spaced uniformly between the top of the fence fabric and the outside barbed wire.

E–5. *Top rails* shall be round (tubular), shall be in lengths not less than 18 feet, and shall be fitted with expansion couplings for connecting the lengths into a continuous run. The couplings shall be not less than 6 inches long, shall provide a substantial connection, and shall allow for expansion and contraction of the rail. Suitable tie clips shall be provided in sufficient number for attaching the fabric securely to the top rail at intervals not exceeding 2 feet. Means shall be provided for attaching the top rail securely to each gate, corner, pull, and end post.

E–6. *Stretcher bars and clips.*—

E–6a. *Stretcher bars* shall be not less than 3/16 by ¾ inch in cross section or equivalent cross-sectional area and shall be of lengths equal to the full height of the fabric with which they are to be used. The stretcher bars shall be arranged for attaching to the fabric by threading through the fabric, by clamps, or by other positive mechanical means. One stretcher bar shall be provided for each gate, pull, and end post, and two for each corner post.

E–6b. *Clips* of adequate strength shall be provided in sufficient number for attaching the fabric and stretcher bars to all posts at intervals not exceeding 15 inches.

E–7. *Top and bottom reinforcing wires.*—Top reinforcing wire shall be provided, when so specified, in lieu of top rail. The reinforcing wires shall be of coiled spring wire not less than 0.177 inch

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± 0.003 inch in diameter. Tie clips shall be provided for attaching each wire to the fabric at intervals not exceeding two feet.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—

F–1a. For the determination of surface defects, dimensions, and weight, at least 2 percent but not less than four samples of each accessory, except reinforcing wire, shall be selected at random for inspection; however, for orders containing four or less of any one or more kinds of accessories, all accessories of such kinds shall be taken.

F–1b. For determination of uniformity and weight of zinc coating, one line post, two line-post tops, two barbed-wire arms, and one top rail shall be taken from each four samples selected under paragraph F–1a, two clips shall be selected from each shipping container, two or more 9-inch specimens shall be taken from each coil of reinforcing wire, and one or more of each size of all other accessories shall be selected at random except that where the number of any one accessory required is less than 15, samples will not be required and acceptance or rejection will be based on tests of similar materials (that is: gates, post braces, and posts other than line posts based on tests of tubular line posts and top rail; stretcher bars, truss rods, post tops, and barbed-wire arms based on tests of line post tops and barbed-wire arms).

F–2. *Inspection.*—The samples shall be examined for surface defects, dimensions, and weights and, for reinforcing wire, each coil shall be gaged for diameter at not less than three points.

F–3. *Zinc-coating test.*—Where material is specified to be given one or both of the special finishes, tests of zinc coating shall be made before the special finish is applied.

F–3a. Uniformity of coating shall be tested in accordance with A.S.T.M. method A 239, Uniformity of Coating by the Preece Test (Copper Sulfate Dip) on Zinc-coated (Galvanized) Iron or Steel Articles, of the issue in effect on the date of invitation for bids. A section may be cut off large accessories for the test.

F–3b. *Rejection.*—If any sample fails to meet any of the test requirements the entire lot represented thereby, including similar accessories of which no samples for test of zinc coating were taken as provided by paragraph F–1b, shall be rejected, except that for accessories rejected on the basis of tests of similar materials, the contractor may elect to have samples selected and tested, the cost of the samples and testing to be paid for by the contractor.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, commercial packaging is acceptable under this specification.

G–2. *Packing.*—Unless otherwise specified, fence posts, gates, top rails, and other members shall be packed for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and afford maximum protection from normal hazards of transportation.

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G–3. *Marking.*—Unless otherwise specified, all shipments shall be marked with the name of the article and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following departmental specifications, of the issue in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments.

H–2. *Army.*—

H–2a. For domestic shipments, current commercial packing which will assure delivery in good order; for export, U. S. Army specification for export packing as specified in the contract or order.

H–2b. U. S. Army Specification No. 100–2, Standard Specification for Marking Shipments by Contractors, copies of which may be obtained without cost upon application to the procuring agency.

H–3. *Navy.*—

H–3a. *Specifications and publications.*—

Navy Department Specification:
General Specifications for Inspection of Material.
Army-Navy Joint Packaging Instruction:
JPI–7—Packaging and Packing of Hardware, Builders and Pole Line.
Bureau of Supplies and Accounts Publication:
Navy Shipment Marking Handbook.

Copies of the specifications and publication listed above may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, New Jersey.

H–3b. *Packaging, packing, and marking.*—

H–3b(1). Fence posts, gates, top rail, post braces, and truss rods shall be packed in conformance with standard commercial practice; other accessories shall be packaged, packed, and marked in conformance with the requirements of Army-Navy Joint Packaging Instruction JPI–7.

H–3b(2). In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

I. NOTES

I–1. *Invitations for bids* should exercise any desired options offered herein and should specify the accessories required, length and number of each kind of post, height of fabric, if barbed wire arms are required, if top rail is not required, if reinforcing wire is required for top or bottom of fabric, width of gate openings, if gates are single or double leaf, fabric for gates if other than that specified herein is desired, if special finishes are desired, requirements

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for lock if required, and requirements for sliding gates if desired in lieu of hinged gates. (See pars. B–1, D–2a, E–1, E–1a, E–1b, E–1e, E–2, E–4, E–4a, E–7, G–1, G–2, G–3.)

I–2. *Erection details* have not been specified. This should be clearly covered in the invitation for bids. For gates, the gate opening, measured as the clear opening distance between inside faces of the gate posts, should be stated. In order that the proper lengths of component members, such as braces and tie rods for corner posts, may be supplied by the contractor, the invitation for bids should state the distance between posts of the fence required. The standard distance between posts is normally not more than 10 feet. Line posts are usually set 3 feet, and other posts 3.5 feet deep. Where barbed wire top is to be used, the gate posts should provide heights approximately one foot above the top of the fabric for attaching the barbed wire.

I–3. Top reinforcing wire is ordinarily used only where the use of top rails is considered unnecessary. The use of bottom reinforcing wire is governed by local conditions, and ordinarily is used only when top reinforcing wire is used.

I–4. The special finishes provided for in paragraph D–2a may be desirable when additional protection against corrosion is needed in locations subject to salt or other extremely corrosive atmospheres, or when other than the zinc finish is desired for appearance or for visibility through the fence.

I–5. The term "corner post" generally refers to a post occurring at any deflection angle of the fence in excess of 10 degrees. Pull posts are usually used to facilitate stretching in long runs of fence and abrupt change of grade.

I–6. Wire fencing fabric is covered by Federal Specification RR–F–191, Fencing; Chain-Link or Welded.

I–7. It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

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